

REVIEW

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Progress in brucellosis immune regulation inflammatory mechanisms and diagnostic advances

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Abstract

Brucellosis, a globally prevalent zoonotic disease caused by *Brucella* species, remains a major public health concern, especially in endemic regions. This study delves into the complex interactions between host immune responses and *Brucella* infection, focusing on both protective immunity and pathological inflammation. Key topics include pathogen recognition pathways, strategies employed by *Brucella* to evade immune defenses, and the collaborative roles of innate and adaptive immunity in driving disease progression. Emphasis is placed on how *Brucella* manipulates host immune regulation to establish chronic infection. From an epidemiological standpoint, this paper discusses transmission routes, risk factors, and the broader implications for global health. Additionally, this study reviews recent advancements in diagnostic techniques, particularly molecular diagnostics and novel biomarker discovery, aimed at enhancing detection accuracy and disease monitoring. Therapeutic strategies, including conventional antimicrobial treatments and emerging immune-based interventions, are also examined, with a focus on their limitations and future potential. This article further addresses challenges such as the heterogeneity of immune responses, obstacles to vaccine development, and the factors contributing to persistent infection. By clarifying the molecular and immunological mechanisms of brucellosis pathogenesis, this work underscores the importance of interdisciplinary approaches to improve understanding, diagnosis, treatment, and clinical management of this enduring infectious disease.

Keyword Brucellosis, Immune response, Chronic infection, Diagnostics, Inflammation, Immunotherapy, Vaccine development

Brucella is a globally prevalent disease caused by bacteria from the *Brucella* family, which is zoonotic in nature and has significant implications for public health and livestock. Despite control efforts in higher-income countries,

brucellosis remains endemic in regions such as the Middle East, Sub-Saharan Africa, South and Central Asia, and parts of Latin America. Brucellosis is a neglected tropical disease according to the World Health Organization, as it tends to be underdiagnosed and causes socioeconomic burden in addition to chronic disease in humans [1]. Human infection with the organism mainly comes about through direct contact with infected animals and consumption of contaminated animal products. Symptoms of brucellosis include an undulant fever along with arthritis and fatigue and neuropsychiatric manifestations. The pathogen's ability to persist within host cells,

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modulate the immune response, and evade clearance complicates diagnosis, treatment, and vaccine development [2, 3]. The immunopathological mechanism of *Brucella* infection is quite complicated and must be well understood for therapeutic innovations and disease control. This review aims to combine new information on the immunology, diagnosis, epidemiology, and treatment of brucellosis with newer technologies and what the future holds for this re-emerging infectious disease (Fig. 1).

Basic immunological mechanisms of brucellosis

Pathogen recognition and immune evasion mechanisms

Brucella species are Gram-negative intracellular bacteria that persist within host cells, causing chronic infections through various immune evasion mechanisms [4]. They modify their lipopolysaccharides (LPS) to reduce TLR4-mediated immune recognition and prevent activation of pro-inflammatory pathways. Once inside macrophages, *Brucella* avoids lysosomal fusion and replicates in specialized vacuoles [5]. By diminishing NADPH oxidase and inducible NO synthase (iNOS), *Brucella* antagonizes the oxidation defense of the host. Consequently, there is a diminished formation of reactive oxygen species (ROS) and nitric oxide (NO). At the same time, it modulates programmed cell death by affecting the Bcl-2 expression and activating MAPK pathways. Disruption of LC3 lipidation and interference with Beclin-1 complex assembly impair autophagic clearance [6]. By enabling an intracellular lifestyle and disabling clearance mechanisms, *Brucella* is able to establish long-term persistence. The sustained intracellular presence leads to low-grade chronic inflammation, which is often associated with granulomatous tissue remodeling, immune exhaustion, and the recurrent symptoms observed in chronic brucellosis such as arthralgia and fatigue. On the adaptive immune front, *Brucella* manipulates TcpB to interfere with TLR signal transduction and NF- κ B activity, thereby blocking antigen presentation and T cell activation [7]. These immune modulation strategies not only enable stealth-like persistence but also offer opportunities for

novel immunotherapeutic approaches, such as nanoparticle vaccines and immune checkpoint modulators currently in early clinical stages [8, 9]. Dysregulation of immune checkpoints like PD-1/PD-L1 and CTLA-4 in chronic brucellosis suggests that immune exhaustion could be targeted therapeutically, with promising results from checkpoint blockade in murine models [10, 11]. In murine models, blockade of these checkpoints has been shown to partially restore T cell responses and reduce bacterial burden, highlighting their potential for adjunctive therapy in refractory cases. Together, these mechanisms enable *Brucella* to establish a persistent, low-profile infection by simultaneously evading detection, blunting effector responses, and inducing immune tolerance. This not only promotes chronic infection but also creates molecular entry points for vaccine and immunotherapy development, particularly in targeting immune checkpoints and enhancing antigen presentation pathways.

Role of the innate immune system in brucellosis

Brucella manipulates the innate immune system to enhance its survival within host cells by promoting immune evasion. It suppresses the pro-inflammatory M1 phenotype while inducing the anti-inflammatory M2 macrophage profile, which aids immune evasion. TcpB disrupts TLR2/4 signaling, pushing macrophages toward an M2 phenotype with reduced bactericidal activity [12]. During the early immune phase, IFN- γ from innate lymphoid cells (ILCs) and NK cells enhances macrophage antimicrobial activity [13], but excessive type I interferons (IFN- α/β) may induce immune tolerance, impairing macrophage activation and bacterial clearance [14]. This interferon imbalance fosters chronic infection, leading to persistent low-level inflammation and symptoms such as joint degeneration and hepatosplenomegaly in chronic brucellosis [15, 16]. Macrophages exhibit functional plasticity, polarizing into M1 or M2 subsets depending on environmental cues. While *Brucella* infection promotes M1 polarization for bacterial clearance, it also drives M2

(See figure on next page.)

Fig. 1 This schematic representation illustrates the pathogenesis of brucellosis, the complexity of host immune responses, and the corresponding clinical management strategies. It depicts multiple transmission routes, such as direct contact, oral ingestion, and occupational exposure, each of which can lead to systemic dissemination and the involvement of various organs, including the brain, lungs, liver, and gastrointestinal tract. The diagram highlights immune system activation, emphasizing the roles of both innate and adaptive immune components. In particular, natural killer (NK) cells, macrophages, and Toll-like receptors (TLRs) are shown to participate in the recognition of *Brucella*, facilitate antigen presentation, and initiate cytokine-mediated signaling cascades. Within the liver, interactions among Kupffer cells, hepatic stellate cells, and infiltrating immune cells contribute to local inflammation and disease progression. Natural killer cells are involved in the detection and elimination of infected host cells by releasing cytotoxic molecules, while macrophages function as antigen-presenting cells that trigger adaptive immune responses through TLR signaling. The figure also presents key diagnostic approaches, including the Rose Bengal Plate Test, polymerase chain reaction assays, and novel biomarker-based techniques, which aim to enhance detection sensitivity and specificity. Lastly, therapeutic strategies are outlined, encompassing conventional antibiotic regimens, immunomodulatory therapies, and emerging personalized treatment modalities

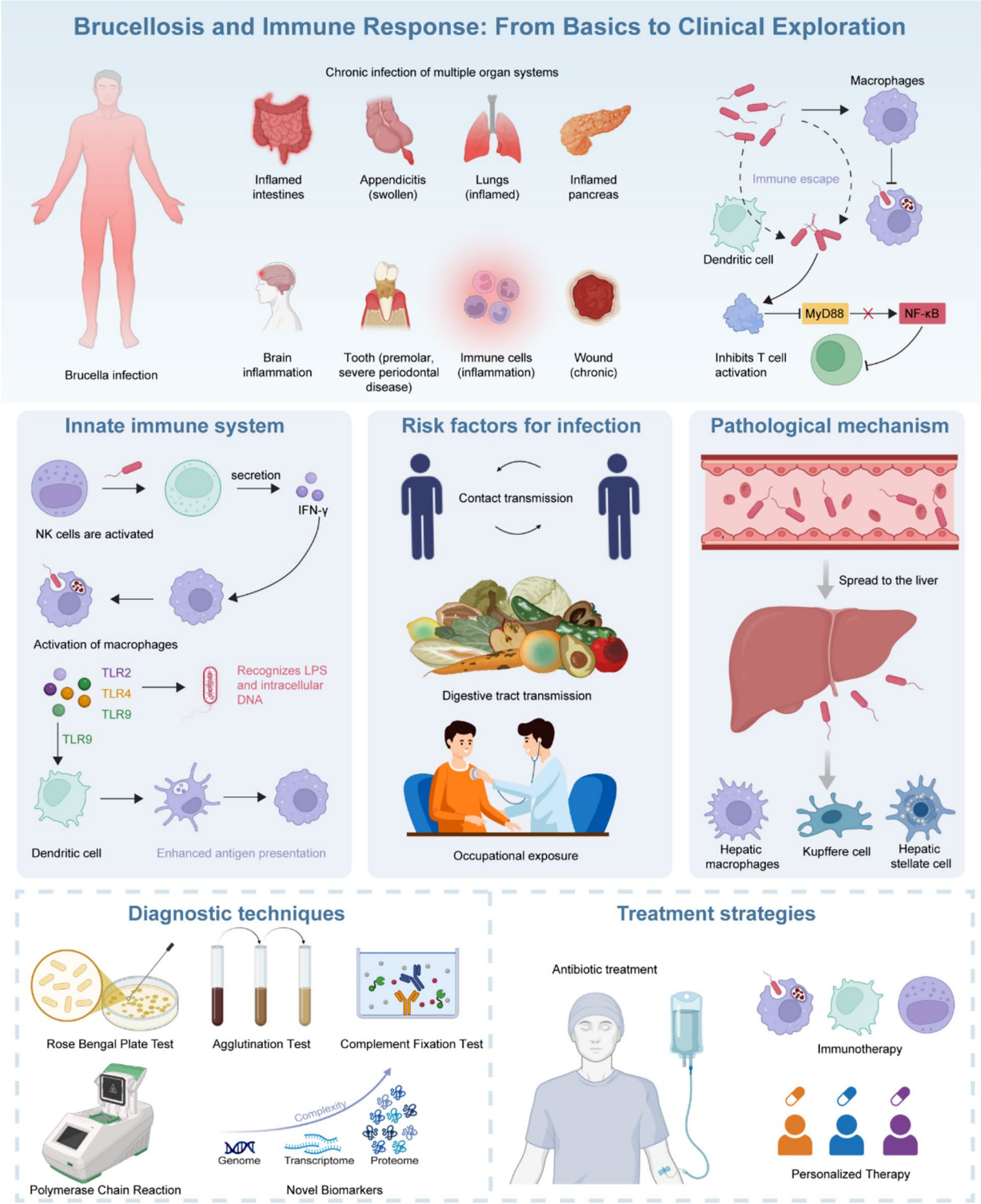


Fig. 1 (See legend on previous page.)

polarization, which weakens immune responses. Chronic brucellosis patients often show an M2-skewed immune response, which correlates with disease persistence and

suboptimal treatment outcomes [17, 18]. Modulating this M1/M2 balance is a potential strategy for improving vaccine efficacy and developing host-directed therapies.

Interaction between adaptive immune responses and brucellosis

Adaptive immunity, particularly Th1 responses, plays a crucial role in controlling *Brucella* infection [19]. CD4⁺ T cells activate macrophages through IFN- γ and TNF- α , while CD8⁺ T cells target infected cells. *Brucella* uses several strategies to evade adaptive immune detection, such as downregulating MHC I and II expression by disrupting antigen presentation pathways through its Type IV secretion system [20]. This leads to a shift in T cell differentiation toward Th2 or Treg profiles, impairing pathogen clearance.

Although primarily intracellular, *Brucella* also induces humoral responses, with B cells producing IgM and IgG to assist phagocytosis, although chronic infection can reduce antibody efficacy [21]. Neutrophils, traditionally part of innate immunity, may influence adaptive responses by competing for antigens or secreting inhibitory factors [22]. Furthermore, immune checkpoint pathways like PD-1/PD-L1 and CTLA-4 are upregulated during chronic *Brucella* infection, contributing to T cell exhaustion and diminished immune responses. Elevated PD-1 expression on T cells reduces cytokine production and proliferation, facilitating immune evasion and persistence. Recent studies suggest that targeting these checkpoints with PD-1/PD-L1 or CTLA-4 inhibitors could restore T cell functionality and improve bacterial clearance, especially when combined with antibiotics. Vaccine trials incorporating checkpoint blockade or new adjuvants show promise in boosting memory T cell responses and long-term protection.

Emerging therapies targeting these immune evasion mechanisms include adjuvant-based vaccines, nanocarriers, and checkpoint inhibitors aimed at enhancing cellular responses and reversing immune suppression [23–25]. New platforms like organ-on-a-chip (OoC) models, such as liver-on-a-chip systems, provide novel ways to study immune–pathogen interactions in a more physiologically relevant context. These systems mimic in vivo tissue architecture, enabling detailed exploration of T cell–macrophage interactions and *Brucella* persistence, offering new insights into chronic infection and immune evasion mechanisms [26, 27]. Integrating such models into brucellosis research could facilitate high-throughput drug screening and the development of immune-modulatory therapies.

Epidemiology and pathophysiological mechanisms of brucellosis

Global epidemiological status of brucellosis

Brucellosis is a widespread zoonosis, particularly in Africa, Asia, and the Middle East [28]. WHO classifies it as a neglected tropical disease. True incidence

is underreported due to poor diagnostics and surveillance. High prevalence areas include Central Asia, North and Sub-Saharan Africa, and parts of Latin America. In China, northern provinces report the highest incidence, though sporadic cases are emerging in the south. Notably, recent years have seen a rising incidence in several emerging regions, such as Southeast Asia and parts of Eastern Europe, where brucellosis was previously sporadic or underrecognized. This trend may be partially attributed to increased international trade in livestock and animal products, informal animal movements, and ecological changes that favor pathogen persistence and transmission. Resurgence in some countries suggests the need for sustained control efforts and international cooperation.

Transmission routes and risk factors of brucellosis

Direct contact, ingestion, and exposure through occupation can lead to transmission [29]. During parturition, milking, or slaughter, animal secretions are primary sources of infection [30]. Consuming unpasteurized dairy or undercooked meat plays a major role [31]. Veterinarians and slaughterhouse workers are considered high-risk professions. The spread of the disease is influenced by soil and water contamination, and animal migration. Further, animals, the interactions among animals, the environment, and humans perpetuate transmission cycles. Both wildlife and domestic animals serve as reservoirs, and contact with livestock and humans facilitates disease transmission. Environmental persistence of *Brucella* spp. Contaminated materials can also indirectly expose humans to toxins. The dynamics of transmission are increasingly driven by emerging risk factors, particularly urbanization and climate change. Urban encroachment into peri-urban areas might enhance human–animal interaction. In addition, climate variability may modify animal migration, environmental survival of pathogens, and vector ecology [32]. The ongoing global spread of brucellosis is driven by globalization as well as environmental and socio-environmental changes [32].

Pathophysiological mechanisms of brucellosis

Brucella boosts its cellular availability and pervasiveness through the modulation of host immune pathways [33, 34]. *Brucella*'s intracellular persistence underlies its chronic pathology [35]. This process is tightly regulated by *Brucella*'s virulence factors such as VirB T4SS, which translocate effector proteins that manipulate host cell trafficking and autophagy pathways. Among organ-specific manifestations, liver involvement is one of the most prevalent and clinically significant. The liver is a key target, where infection leads to HSC activation, collagen deposition, and fibrosis. It invades and replicates within

macrophages by forming a replicative niche known as the *Brucella*-containing vacuole, which avoids lysosomal fusion and instead traffics along the endoplasmic reticulum pathway. Skeletal involvement is another critical organ-specific pathology. In bones and joints, TLR2-mediated cytokine release promotes osteoclastogenesis, leading to bone resorption and joint destruction. Clinical case series have reported that sacroiliitis and spondylodiscitis are among the most common osteoarticular complications, often requiring prolonged antibiotic therapy and sometimes surgical intervention. During pregnancy, *Brucella* infects the placenta, triggering intense local inflammation and immune dysregulation. This placental colonization compromises trophoblast function and vascular remodeling, significantly increasing miscarriage risk [36]. These systemic and localized effects underscore the pathogen's capacity to cause multi-organ damage and establish chronic, relapsing disease.

Advances in diagnostic techniques for brucellosis

Traditional diagnostic methods and their limitations

Brucellosis is a zoonotic disease with non-specific symptoms like fever, fatigue, and joint pain [37]. Traditional diagnostic methods, including serological tests and bacterial culture, are commonly used but have notable limitations. Serological tests such as RBPT and SAT are widely employed but suffer from low sensitivity and specificity, often resulting in false positives due to cross-reactivity with other bacteria. Blood cultures, considered the gold standard, are slow growing, taking up to 3 weeks, and have low positivity rates [38]. These drawbacks highlight the need for more sensitive, rapid, and specific diagnostic approaches.

Application of molecular diagnostic techniques in brucellosis

Molecular techniques like PCR, qPCR, and LAMP have significantly improved brucellosis diagnosis by enabling faster and more accurate detection [39–42]. PCR targets specific *Brucella* genes, offering high sensitivity and specificity [43]. LAMP delivers results more quickly than thermal cyclers and has also demonstrated excellent sensitivity and specificity [44]. Additionally, proteomics using the VirB protein enhances diagnostic accuracy further [45]. However, PCR's diagnostic performance can vary across patient populations, and variability in primer efficiency can affect amplification, impacting precision. Furthermore, the high cost of qPCR equipment and reagents limits its accessibility, particularly in resource-constrained regions, where even basic infrastructure may be lacking. LAMP offers an equipment-independent option but requires careful

primer design to avoid non-specific amplification. Both PCR and qPCR methods are susceptible to contamination, leading to false positives if strict laboratory protocols are not followed. Thus, clinical validation in diverse patient populations is essential for reliable results.

Potential of novel biomarkers in the diagnosis of brucellosis

Emerging omics technologies are helping identify novel biomarkers for brucellosis [46, 47]. Non-coding RNAs, especially piRNAs and miRNAs, hold potential as disease markers. PiRNAs offer disease-specific patterns, while miRNAs play a role in the inflammatory response [48]. Proteomics has identified proteins like VirB12, which show better diagnostic sensitivity and specificity than traditional methods [49]. Metabolomics, by detecting metabolites altered during infection, also offers promise. However, large-scale application of these methods is hindered by the need for biomarker validation and standardization of detection methods. Further clinical studies are needed to assess the sensitivity and specificity of these novel biomarkers across different populations. For example, piRNAs have shown potential in distinguishing between active and past infections. While these early findings are promising, more clinical evidence is necessary before piRNAs can be adopted for routine diagnostics. Combining biomarkers could improve diagnostic accuracy and clinical utility [50]. However, new technologies still face challenges such as high costs, limited accessibility, and the need for specialized equipment, which may restrict their use in clinical settings. Currently, there are no effective markers to assess treatment efficacy. Small titer changes in the Standard Agglutination Test (SAT) guide treatment decisions, but they may not accurately reflect bacterial clearance. This can lead to incomplete or unnecessarily prolonged treatments, raising the risk of drug resistance and adverse effects. Differentiating between relapse and reinfection is another challenge. In endemic areas, patients can remain seropositive for months or even years, complicating the diagnosis of whether a patient with recurring symptoms is experiencing a true relapse or has acquired a new infection. This uncertainty increases the likelihood of misdiagnosis and inappropriate treatment. Therefore, there is an urgent need for dynamic biomarkers that can reflect disease activity, distinguish between active and past infections, and differentiate *Brucella* relapse from other causes. These biomarkers must undergo rigorous clinical validation to ensure they are reliable and suitable for routine use in clinical practice.

Treatment strategies and clinical practice in brucellosis

Current status and challenges of antibiotic treatment for brucellosis

Antibiotic therapy is the primary treatment for brucellosis, but due to *Brucella*'s ability to survive inside host cells, single antibiotic treatments often fail [51]. *Brucella* evades antibiotics by residing within macrophages, where it is shielded from both the immune system and many antibiotics. The standard treatment is combination therapy, usually doxycycline with rifampicin for at least six weeks (Fig. 2). In severe cases, aminoglycosides like streptomycin may be added. However, treatment failures and relapses are common, especially in endemic regions. Recurrence, marked by the return of symptoms and re-isolation of *Brucella*, is often due to its persistence within intracellular reservoirs that resist immune clearance and antibiotic action. The formation of biofilms by *Brucella*

further complicates treatment, alongside factors like bacterial biotype changes, antibiotic resistance, and host immunity. Additionally, *Brucella* has mechanisms like efflux pumps and target site mutations that contribute to antibiotic resistance. Clinical management is complicated by non-specific and fluctuating symptoms, often leading to delays in correct diagnosis and treatment, especially in cases resembling rheumatic diseases [52].

Recent studies suggest that triple therapy could reduce recurrence, although risks like drug toxicity and resistance development remain [53]. New antibiotics, such as tigecycline and fluoroquinolones, have shown in vitro effectiveness against *Brucella*, though clinical validation is limited [54, 55]. These antibiotics, in combination, may offer better intracellular penetration and broader efficacy. Host-directed therapies, aimed at modulating the immune system to aid in pathogen clearance, are also being explored. Alongside these, innovative approaches

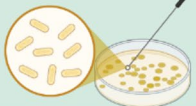
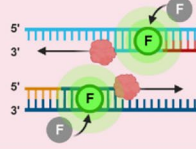
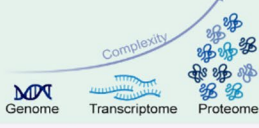
Diagnosis/Treatment	Key Approaches/Regimens	Advantages and Challenges
Traditional Diagnosis 	a) Serological Tests (Rose Bengal Plate Test [RBPT], Standard Agglutination Test [SAT], Complement Fixation Test [CFT]) b) Bacterial Culture (blood, bone marrow)	a) Low cost and simple operation b) Culture remains the gold standard c) Low sensitivity, poor specificity and time-consuming
Molecular Diagnosis 	a) PCR (target genes: 16S rRNA, bcp31, etc.) b) qPCR (quantitative analysis) c) LAMP (isothermal amplification)	a) Rapid (hours), high sensitivity (92.3%) and specificity (99.3%) b) LAMP suitable for field use c) Equipment dependency (e.g., qPCR), technical expertise required; LAMP requires optimized primer design
Novel Biomarkers 	a) Non-coding RNAs (piRNA, miRNA) b) Proteomics (VirB proteins, differentially expressed proteins) c) Metabolomics (amino acid metabolites)	a) Non-invasive, high specificity (e.g., VirB12: 94%) b) Multi-marker panels enhance accuracy c) Insufficient clinical validation, lack of standardized protocols; and requires large-scale multicenter studies
Antibiotic Therapy 	a) Dual Therapy (doxycycline + rifampicin) b) Triple Therapy (+ streptomycin/gentamicin/ fluoroquinolone)	a) Standardized regimens; triple therapy reduces relapse rates b) Rising drug resistance c) Toxicity risks (nephro-/ototoxicity), prolonged duration
Immunotherapy 	a) Phage lysates (animal models) b) Subunit Vaccines (OMP31, OMP25) c) Cytokine Therapy (IFN-γ)	a) Enhances host immunity; vaccines may prevent recurrence b) No approved human vaccines c) Safety/efficacy unverified; limited clinical translation
Comprehensive & Personalized Therapy 	a) Surgical Intervention (e.g., spinal stabilization, valve replacement) b) Multidisciplinary Team (MDT) c) Personalized Adjustments (pediatric, pregnant, elderly)	a) Improves cure rates in severe cases (e.g., endocarditis: 90%) b) Reduces relapse (spondylitis: 11%) c) Requires multidisciplinary coordination; complex case management; need for dynamic toxicity monitoring

Fig. 2 Diagnosis and therapy of Brucellosis

such as nanoparticle-based drug delivery systems may enhance antibiotic delivery to infected cells, reducing toxicity and improving efficacy [56–62]. For example, encapsulating rifampicin or doxycycline in liposomes or nanoparticles can increase intracellular concentrations and decrease relapse rates in experimental models [63, 64]. Furthermore, combining nanoparticle-loaded antibiotics with immune-modulating agents like IFN- γ has shown promise in promoting bacterial clearance [65, 66]. However, challenges related to stability, delivery, and immune responses need to be addressed before clinical use. Phage therapy, still experimental, has shown potential as an alternative. Bacteriophages targeting *Brucella* exhibit bactericidal activity both in vitro and in infected macrophages [67, 68]. When combined with subtherapeutic doses of antibiotics, phage-antibiotic synergy enhances bacterial killing and reduces the risk of resistance development. Additionally, understanding host–pathogen interactions, especially the genetic factors influencing treatment response, could lead to more personalized therapeutic strategies [69]. Genetic markers related to immune response, such as polymorphisms in TLRs or cytokine receptors, could influence infection clearance and antibiotic efficacy [70].

Progress in immunotherapy for brucellosis

Immunotherapy, still in the exploratory stage, has shown potential in treating brucellosis (Fig. 2) [71]. Since *Brucella* evades immune clearance, enhancing the host's immune response is crucial. Animal studies have shown that bacteriophage lysates can stimulate strong immune responses, significantly reducing *Brucella* in infected cattle. Recent murine model studies have also demonstrated that nanoparticle-based subunit vaccines promote Th1-mediated responses, aiding bacterial clearance and reducing spleen colonization [72]. While no human vaccine is approved, attenuated live vaccines for animals have shown strong immunogenicity, though they carry risks like abortion [73]. Subunit and DNA vaccines are being researched, with promising results indicating safety and immune protection. A recent phase I clinical trial of a DNA vaccine targeting *Brucella* outer membrane proteins (OMPs) showed favorable safety and immunogenicity profiles in healthy volunteers, marking progress toward human applications [74]. Multi-antigen vaccine designs could further optimize antigen coverage, offering a new direction for brucellosis vaccine development [75]. Cytokine-based therapies, like exogenous IFN- γ , have also shown potential but require further human trials [76]. T cell therapies could offer a strategy to modulate immunity in chronic infections [77]. However, immunotherapy research is limited by the lack of robust clinical trials

and long-term safety data. Future studies should integrate data from preclinical and early-phase human trials to bridge the gap between research and clinical application. Additionally, integrating host immunogenetic profiling could enhance vaccine efficacy and predict immune responses.

Integrated and personalized treatment approaches for brucellosis

As brucellosis treatment becomes more complex, integrated and personalized approaches are becoming essential [78]. Integrated treatment combines antibiotics, immune modulation, and, when necessary, surgery, tailored to disease severity and host immune status [79]. For severe cases like spondylitis or infective endocarditis, multi-drug regimens combined with surgical interventions have improved outcomes. Personalized treatments consider factors like drug metabolism and comorbidities, particularly in children, pregnant women, and the elderly [80]. The WHO recommends sulfamethoxazole trimethoprim plus rifampin for children, and rifampin with ceftriaxone for pregnant women. Multidisciplinary team (MDT) management has proven effective for complex brucellosis cases, especially those involving bone, joint, or organ involvement, resulting in higher cure rates and more tailored treatment plans. The future of brucellosis management lies in advancing personalized approaches and MDTs [81]. Additionally, the role of stem cell exosomes in tissue repair suggests potential for treating bone and joint injuries caused by brucellosis [82]. Future research should explore how genomic, metabolomic, and immunologic data can further personalize therapy and improve prognosis. Bacteriophage-based therapies or microbiota modulation also represent promising, though underexplored, strategies for managing refractory brucellosis.

Post-treatment management and long-term outcomes

Despite appropriate antibiotic therapy, many brucellosis patients experience relapses or lingering symptoms, such as fatigue, arthralgia, and depression [83]. These symptoms may persist for months or years, severely affecting quality of life. Long-term management includes regular clinical follow-ups, monitoring inflammatory markers, and supportive treatments like physiotherapy or psychological counseling when needed [84]. Standardizing post-treatment care protocols could improve prognosis and reduce relapse rates, particularly in high-risk populations. Prospective cohort studies are necessary to better understand the natural history of post-treatment brucellosis and identify predictive biomarkers for recurrence.

Controversial issues in the immunological research of brucellosis

Individual variability in the immune response to brucellosis

The clinical diversity observed in brucellosis can be attributed to significant variations in host immune responses [85]. These variations encompass both humoral and cellular immune responses, cytokine profiles, and immune cell activities, which are influenced by genetic factors and prior infections. Advanced techniques such as transcriptomics and single-cell sequencing have begun to shed light on the mechanisms behind this variability [86–89]. *Brucella* infection typically triggers humoral immunity characterized by IgG1 and IgG3 antibodies, yet there is considerable inter-individual variation in antigen recognition patterns [90]. However, despite the production of specific antibodies, humoral responses alone are often inadequate for clearing *Brucella* infection, likely due to the bacterium's ability to persist within intracellular compartments, thus evading antibody-mediated neutralization [8]. *Brucella* employs multiple immune evasion strategies, including the inhibition of phagosome–lysosome fusion, suppression of host cell apoptosis, and modulation of innate immune signaling pathways such as NF- κ B and type I interferons. These mechanisms enable *Brucella* to maintain a chronic intracellular lifestyle, which impedes the activation of effective immune clearance. Cell-mediated immunity, particularly the Th1-mediated production of IFN- γ and TNF- α , plays a critical role in protective immunity against *Brucella*. However, significant immunological heterogeneity is observed, with some individuals exhibiting dominant Th2 responses or increased regulatory T cell activity [91]. Such immune deviations may hinder effective pathogen clearance, as Th2 or Treg-dominated responses suppress the pro-inflammatory mechanisms essential for bacterial eradication. Variations in innate immune factors, such as differences in Toll-like receptor (TLR) expression and downstream signaling, further modulate inflammation and shape subsequent adaptive immune responses [92]. Host-specific factors, including genetic background and mucosal immunity integrity, also influence vaccine responsiveness, highlighting the need for personalized therapeutic strategies in the management and prevention of brucellosis [93].

Challenges and breakthroughs in brucellosis vaccine development

Vaccination is a key tool in brucellosis prevention; however, ongoing vaccine development faces numerous challenges [94]. The sophisticated immune evasion mechanisms of *Brucella*, combined with its antigenic diversity,

complicate the design of vaccines that provide consistent and robust protection [95]. The Type IV secretion system and cyclic β -1,2-glucans are pivotal in *Brucella*'s subversion of host immunity and present potential targets for rational vaccine design [96]. Currently available veterinary vaccines, such as Rev1, S19, and RB51, have shown effectiveness in reducing transmission. However, they are associated with significant biosafety concerns, including the risk of abortion and reproductive complications in pregnant animals [97]. Additionally, *Brucella*'s ability to manipulate immune checkpoints—such as PD-1/PD-L1 and CTLA-4—poses a substantial challenge for vaccine efficacy. These immune checkpoint pathways may suppress adaptive immune responses to vaccination, thus limiting the effectiveness of existing vaccine candidates. Recent studies suggest that combining vaccines with immune checkpoint inhibitors may enhance immune responses, overcoming the immune tolerance induced by *Brucella*'s stealth mechanisms [46]. However, live attenuated vaccines carry risks, including the potential for reversion to virulence and interference with serological diagnosis, complicating surveillance efforts.

The considerable genetic variation among *Brucella* strains further complicates the development of a universally effective vaccine [98]. The evolving nature of *Brucella* strains and their immune evasion tactics indicate that a single vaccine may not suffice for all strains, presenting a major challenge for global vaccine distribution. Emerging research into subunit and DNA vaccines, which target specific *Brucella* antigens, has shown promise, but success in human trials has been limited. For example, the L7/L12 ribosomal protein-based vaccine and Omp31 subunit vaccines have demonstrated immunogenicity in animal models but failed to provide adequate protection in phase I/II human trials, often due to insufficient T cell activation or declining antibody titers [99]. Adverse reactions in clinical trials, including transient fever, injection site inflammation, and fatigue—especially in platforms using viral vectors or potent adjuvants—pose challenges for vaccine acceptance and regulatory approval, particularly for large-scale immunization programs [100]. Emerging adjuvant technologies, such as chitosan nanoparticles or TLR agonists, are being explored to improve immune responses and reduce immune evasion. However, these approaches face limitations, including poor immunogenicity, the need for repeated boosters, and a lack of durable immune memory, which restrict their applicability in endemic regions. A promising approach involves combining vaccines with immune checkpoint inhibitors or cytokine modulation to overcome immune suppression and enhance vaccine efficacy [101]. Nonetheless, further preclinical models and

well-designed clinical trials are needed to validate the safety and long-term efficacy of these novel vaccine platforms. Given the burden of chronic brucellosis and the role of immune evasion in disease persistence, next-generation vaccines must address immune tolerance and exhaustion. Vaccine platforms that induce robust T cell responses while bypassing *Brucella*'s immune evasion strategies could offer a more effective solution for brucellosis prevention and treatment.

Mechanisms of chronicity and immunological sequelae in brucellosis

Chronic brucellosis can persist even after appropriate treatment, with symptoms such as fatigue, joint pain, and depression lingering long after the acute infection phase (Fig. 2) [102]. This persistence is thought to result from continued immune activation, pathogen survival within immune cells, and immune reprogramming. *Brucella* can remain latent within macrophages and dendritic cells, causing a persistent low-grade inflammatory response (Fig. 2) [103].

Additionally, *Brucella* employs molecular strategies, such as upregulating virulence factors, to manipulate host cell signaling pathways and favor intracellular persistence [96]. These mechanisms enable *Brucella* to alter antigen presentation and inhibit apoptosis, thereby promoting chronic infection.

Chronic brucellosis is also associated with immune tolerance, with *Brucella* inducing immune-suppressive factors, such as IL-10 and TGF- β , which limit the host's ability to clear the infection [104]. Recent studies suggest that chronic infection is linked to dysregulated immune checkpoint expression (PD-1, CTLA-4) on T cells, as well as impaired dendritic cell maturation, which together sustain an immunosuppressive environment [11]. T cell exhaustion and the establishment of immune memory at tissue sites may contribute to the recurrence and persistence of symptoms.

Emerging evidence points to the neuroimmune environment as a key factor in chronic brucellosis. Infected individuals often experience cognitive and mood alterations, which may be linked to inflammatory responses in the central nervous system (CNS) [105]. *Brucella* can disrupt the blood–brain barrier and activate microglial cells, leading to sustained production of inflammatory mediators, such as IL-6 and TNF- α , which contribute to neurodegenerative-like changes [106]. The altered communication between the nervous and immune systems may elucidate the fatigue observed in chronic brucellosis. These findings underscore the importance of integrating neuroimmunology and chronic inflammation studies into our understanding of brucellosis pathogenesis.

Future research directions and prospects for brucellosis

Prospects for multidisciplinary collaborative research on brucellosis

The control and prevention of brucellosis is a multifaceted challenge that requires a multidisciplinary approach [107, 108]. As a disease that spans across medicine, veterinary science, microbiology, immunology, ecology, epidemiology, public health policy, and socioeconomics, brucellosis necessitates collaboration from various fields. The widely adopted "One Health" concept is fostering a more integrated approach to brucellosis prevention and control [109]. The effectiveness of multidisciplinary cooperation has been demonstrated at the practical level in several countries [110]. For instance, in India, the establishment of a brucellosis prevention and control system has facilitated interdisciplinary expert discussions on critical issues, such as the development of data-sharing platforms, improved interdepartmental communication, and enhanced epidemiological studies, thus enabling more localized and targeted control efforts. In developed regions, such as Europe and North America, joint veterinary public health systems and regional animal disease monitoring networks have proven effective in early detection, outbreak management, and vaccination efforts for brucellosis.

Multidisciplinary collaboration not only enhances the practical application of brucellosis control measures but also provides fresh perspectives on disease mechanisms in basic research [111]. For example, combining molecular microbiology and host immunology has led to the discovery that *Brucella* uses its Type IV secretion system to inject effector molecules into host cells, thereby interfering with immune signaling and evading immune responses [112, 113]. This breakthrough has identified molecular markers that could aid in the development of more targeted vaccines or treatment strategies. Additionally, the application of systems epidemiology and ecological modeling has visualized the transmission pathways of *Brucella* in complex ecosystems, offering a scientific foundation for the creation of early warning systems and identification of high-risk areas [114]. The technological scope of brucellosis research has been significantly enhanced by advancements in materials science, bioinformatics, nanotechnology, and artificial intelligence [115–120]. For example, the combination of nanomaterials and biosensors has enabled the creation of rapid, ultra-sensitive diagnostic devices [121–124]. Artificial intelligence (AI) is now being used to predict *Brucella* transmission patterns, establish early warning systems, and assist in personalized treatment decisions, all of which improve diagnostic accuracy [125, 126]. Bioinformatics tools help researchers analyze *Brucella* genomic

data, which in turn provides insights into host–pathogen interactions and forms the molecular basis for developing targeted vaccines and treatment strategies [127, 128]. Furthermore, new antibacterial hydrogels, such as photo-dynamic hydrogels and ultrasound-responsive hydrogels, are showing promise in enhancing antimicrobial efficacy and providing innovative solutions for localized management of brucellosis-related wounds [129, 130]. Thus, the future of brucellosis research and control will depend on strengthening interdisciplinary collaboration and building a more integrated, multidimensional research framework to address the disease.

Potential applications of emerging technologies in brucellosis research

The advent of cutting-edge technologies such as biosensors, proteomics, next-generation sequencing (NGS), CRISPR gene editing, and AI has revolutionized brucellosis research, particularly in the areas of diagnosis, treatment, and vaccine development. These innovations are driving brucellosis research into a new era, one that is increasingly technology driven. However, while promising, these technologies come with limitations. Issues such as cost, accessibility, technical complexity, and integration into existing healthcare infrastructures remain significant barriers to their widespread adoption. Biosensor technology, for instance, has shown great potential for the rapid diagnosis of brucellosis [131, 132]. Traditional diagnostic methods, such as the Rose Bengal test and ELISA, are simple but lack the sensitivity and specificity required for emergency field testing. Recent advancements in immunosensors utilizing quantum dots (QDs) and graphene oxide nanomaterials have demonstrated ultra-high sensitivity, portability, and real-time operation [133]. Despite these advancements, biosensors still face challenges, including complex manufacturing processes and sensitivity to environmental factors, such as humidity and temperature, which may limit their applicability in resource-poor settings.

Proteomics technology has significantly aided in the identification of novel diagnostic biomarkers and vaccine targets [134, 135]. By using isotope-labeled quantitative proteomics techniques, such as TMT and iTRAQ, researchers can perform comprehensive analyses of protein expression changes during infection [136]. For example, the VirB protein has been identified as a highly expressed protein across various host models, with good antigenicity and species conservation, making it a promising target for serological diagnosis [137]. These proteins, which play crucial roles in immune response regulation, are also potential candidates for subunit vaccine development [138]. However, proteomic analyses generate vast amounts of data, requiring sophisticated

bioinformatics tools to distinguish meaningful biomarkers from background noise. Additionally, most findings are still in preclinical stages, and further validation is needed to translate these discoveries into reliable diagnostic tools.

Despite significant advances in identifying vaccine targets, there is an ongoing need for more effective human vaccines. Current veterinary vaccines, such as Rev1, S19, and RB51, have proven beneficial but show limited efficacy in humans and pose safety concerns, such as the risk of abortion in pregnant women [139]. Moreover, the immunity provided by these vaccines is not long lasting, further complicating the challenge of immune evasion by *Brucella* [95]. Thus, the development of vaccines targeting conserved and immunogenic antigenic profiles remains a critical area of research [140, 141]. NGS has significantly improved brucellosis diagnostics, especially in chronic and atypical cases [142]. NGS offers high-throughput capacity and superior sensitivity that extends beyond the limitations of conventional diagnostic methods for pathogen detection. In cases of osteoarticular brucellosis, where common culture techniques often fail to produce results, directly assessing the lesion via NGS (next-generation sequencing) can aid in the identification of the species and guide clinical decisions effectively. The high cost involved, requirement of fancy machines, and need for specialists to analyze the data are hindering the implementation of NGS in practical clinics. Also, the lack of standardized sample processing and interpretative protocols in brucellosis is missing.

Gene editing technologies, especially CRISPR-Cas systems, have significantly advanced functional genetic studies on *Brucella* [143–146]. Researchers now use these tools to create gene knockout or overexpression mutants to identify virulence factors and understand immune evasion mechanisms, thereby enhancing our understanding of *Brucella* pathogenicity. Advances in synthetic biology for vaccine development also involve producing modified vectors or engineered bacterial strains that express targeted antigens to improve immunogenicity and reduce toxicity [147]. However, regulatory and safety concerns remain regarding the use of engineered strains, and off-target effects in gene editing experiments require ongoing monitoring. AI and machine learning have the potential to transform brucellosis research, extending to imaging diagnostics, epidemic forecasting, and vaccine optimization [148, 149]. Data platforms that combine global infection databases, vaccination records, and animal movement tracking enable predictive modeling of disease spread, which supports evidence-based, region-specific prevention strategies. While AI-driven solutions offer tremendous promise, their effectiveness is highly dependent on data

quality and representativeness. In resource-poor areas, where surveillance data may be sparse or outdated, the accuracy of AI predictions could be compromised.

Global strategies and policy recommendations for brucellosis prevention and control

Effective global control of brucellosis requires comprehensive strategies supported by robust policy frameworks [150]. Livestock vaccination remains the cornerstone of brucellosis prevention, as demonstrated by regional mass immunization programs that significantly reduce disease incidence and limit outbreak risks [151]. Consequently, global brucellosis control policies should prioritize the implementation of animal vaccination campaigns. Equally crucial are enhanced epidemic surveillance and early warning systems. The development of integrated “One Health” monitoring platforms, using human, animal and environmental health data, can improve the accuracy and timeliness of surveillance, enhance early outbreak detection and enable efficient epidemic containment.[152]. Government funding and policy support are critical for the effective and sustained implementation of brucellosis control measures. Authorities must allocate additional funding for vaccine development, diagnostic technology, and preventive interventions [153]. Strengthening legal frameworks governing animal trade and farming practices through dedicated legislation is also essential to minimize transmission risks [154]. International collaboration and knowledge exchange are vital for global brucellosis control [155]. Given the transboundary nature of the disease, enhancing multinational cooperation through joint research initiatives, standardized diagnostic protocols, shared surveillance databases, and harmonized vaccination strategies is essential. Cross-border training programs for veterinarians and public health professionals, collaborative outbreak investigations, and coordinated response planning can significantly improve global control efforts [156]. Future research should focus on overcoming current limitations in diagnostics, addressing barriers to vaccine development, and ensuring the equitable application of emerging technologies across both developed and developing regions. Integrating AI, NGS, and gene editing technologies into accessible public health systems should be a top priority. Longitudinal studies to evaluate vaccine efficacy, immune response variability, and environmental factors influencing *Brucella* transmission are also critical. Ultimately, effective brucellosis control requires a globally coordinated, forward-looking research agenda that fosters technological

innovation and supports the development of adaptable, sustainable health systems worldwide.

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Author contributions

PP.L. wrote the main manuscript text, contributed to the design and methodology of the study, performed the data collection and interpretation. B.L., Q.W. and LL.W. contributed to the design and methodology of the study. MS.C. and JT.J. performed the data collection and interpretation. M.Y. and JY.P. assisted with the data collection. J.L. and ZW.L. provided overall supervision and guidance throughout the study. All authors reviewed and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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